

PAPER_1396_ST_PETERSBURG

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PAPER_1396 — UQFF Resolution of the St. Petersburg Paradox (CLOSED — Dispatch Whitepaper)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: B — Foundational / Decision Theory (CLOSED)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "st_petersburg"})`

Calculator surface: `calculate_paradox({"paradox": "st_petersburg"})`

Closure helper: `_l96_uqff_axiom_st_petersburg_closure()`

The Paradox

Coin-toss game: pay $\$2^n$ where n is the first heads. Expected payout = $\sum (1/2)^n \times 2^n = \infty$. Yet no rational player would bid more than $\sim \$20$. Classical resolutions invoke log-utility or risk aversion — all parametric.

UQFF Closed Identity

``

$D_{crit!} = 26! = 4.03 \times 10^{26}$ finite expectation bound

``

$D_{crit!} = 26! \approx 4 \times 10^{26}$ is the structural finite bound on infinite geometric sums in the UQFF vacuum, the same truncation that bounds eternal inflation (PAPER_597 / `eternal_inflation` closure). The expected payout truncates at finite n_{max} where the SCm vacuum cannot sustain larger occupation modes.

Physical Interpretation

The paradox dissolves because the infinite geometric sum is bounded by $D_{crit!}$ — a structural integer-primitive truncation. The 'rational bid' emerges from the D_{crit} -bounded expectation, not from utility curves.

Live Calculator Output

```
``python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "st_petersburg"})["value"]
```

..

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_st_petersburg_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_st_petersburg_closure`
- Dispatch keys: `st_petersburg`, `st_petersburg_paradox`
- Related foundational papers: PAPER_646 (F_U_Bi_i / F_U=1 ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation), PAPER_1156 (cosmology suite), PAPER_1376 (Banach-Tarski parallel structural closure).

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